Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/24/2004
PI Name:	Matalon, Moshe Ph.D.		
Project Title:	Modeling Microgravity Non-Premixed Combustion Systems		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Evanston	**State:**	IL
Zip Code:	60208-3125	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	11/16/2000	**End Date:**	11/15/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2511		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The task involves identifying and analyzing relatively simple models of combustion of non-premixed systems. In particular, the task is to describe the development of spontaneous flame oscillations, the mechanisms responsible for such behaviors and the nature of the oscillations. Another task is to describe the onset of other form of instabilities, such as cellular structure, and how, in particular, they are affected by the flow field and the reactants compositions. The objective of this work is to provide theoretical support to microgravity experimental studies of non-premixed combustion (diffusion flames) in an attempt to develop rigorous understanding of the effects of gravity, or the lack of gravity, on these processes. More specifically, the objective is to determine the conditions for the onset of flame instabilities, and identify the responsible physical mechanisms that lead to these modes of burning. Many practical applications and most fires consist of diffusion flames. Understanding fires is particularly important to NASA's missions because of its concern with the safety of future space explorations.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the current project focuses on configurations most relevant to microgravity studies, the results being fundamantal in nature are also relevant to to the understanding of the structure and dynamics of diffusion flames which are most relevant to many earth applications. Many practical applications of combustion, such as furnaces and diesel engines, involve diffusion flames and in most cases fires result from the combustion of non-premixed reactants. Better understanding leads to imporvement in design, combustion efficiency and indirectly reduction of polutants.
Task Progress:	Progress has been made on the dynamics of edge-flames in mixing layers. The objective was to systematically identify the conditions for the onset of oscillations examining, in particular, the influence of differential, mixture-strength, flow rate and radiative heat losses. We have continued our work on the stability of diffusion flames and have succesfully designed a burner for the realization of an unstrained one-dimensional diffusion flame. Planar flames are observed at a standoff distance that compares well with the results of our theoretical model and the onset of cellular structures appears to agree with the theoretical predictions.
Bibliography Type:	Description: (Last Updated: 04/27/2010)
Articles in Peer-reviewed Journals	Antoniou, E.S.; Bechtold; J.K.; Matalon, M. "A Diffusional-Thermal Theory of Near-Stoichiometric Premixed Flames" SIAM Journal on Applied Mathematics 64(4), pp.1434-1456 , May-2004
Articles in Peer-reviewed Journals	Armendariz, J.; Matalon, M.. "Marangoni Instability at a Liquid-Vapor Interface of a Burning Thin Film" Physics of Fluids 15(5):1122-1130 , Jan-2004
Articles in Peer-reviewed Journals	Cui C.; Bechtold J.; Matalon M. "Hydrodynamic Theory of Premixed Flames: Effects of Stoichiometry, Variable Transport Coefficients and Arbitrary Reaction Orders " Journal of Fluid Mechanics 487, pp. 179-210,2003 , Jan-2004
Articles in Peer-reviewed Journals	Cui C, Matalon M, Daou J, Dold J. "Effects of differential diffusion on thin and thick flames propagating in channels." Combustion Theory and Modelling 2003 Dec;8(1):41-64. http://dx.doi.org/10.1088/1364-7830/8/1/003 , Jan-2004
Papers from Meeting Proceedings	Kurdyumov, V.; Matalon, M. "An Oscillating Edge-Flame in a Mixing Layer" American Institute of Aeronautics and Astronautics AIAA Paper 2003-1016 , Jan-2004
Papers from Meeting Proceedings	Cui C.; Matalon, M.; Jackson, T.L. "Flame Propagation in Channels: Steady and Pulsating propagation" American Institute of Aeronautics and Astronautics, Reston VA AIAA Paper 2004-298 , Jan-2004